

G code

- Gcode is how we tell our machines what they need to do.
- It's been from around late 1950.
- widely Used CNC
Computer Numerical Control

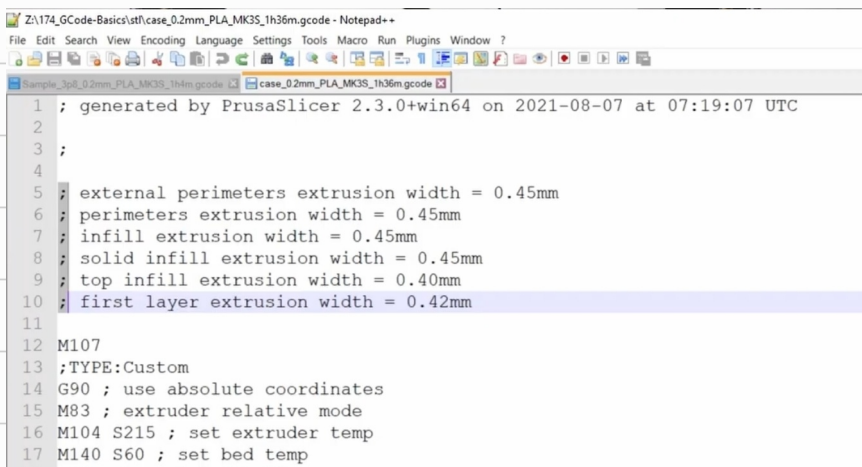
- File structure

- 1) Header : which is general information.
- 2) Start Gcode : which we can customise in our slicer.
- 3) Printing Commands
- 4) End Gcode

There also often block of text that contains settings that are used.

→ it is really good, as some profiles really worked good.

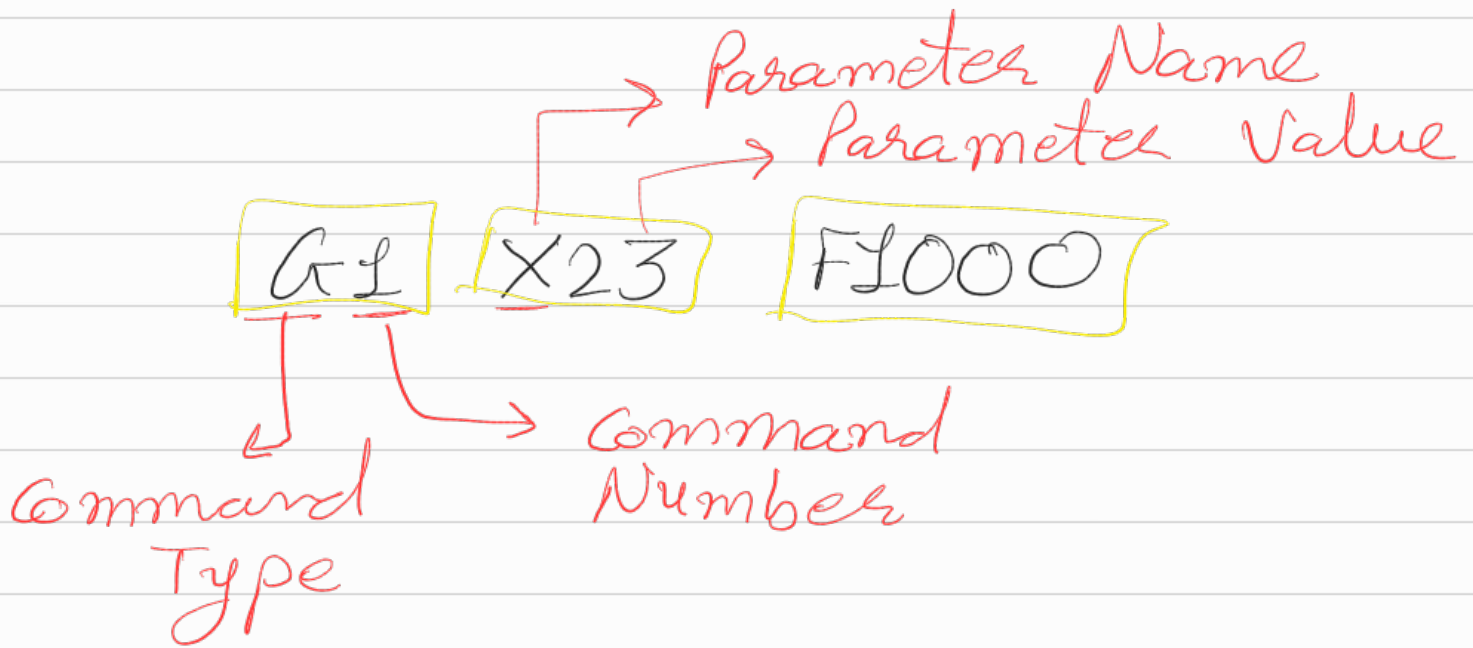
1) Header



```
1 ; generated by PrusaSlicer 2.3.0+win64 on 2021-08-07 at 07:19:07 UTC
2
3 ;
4
5 ; external perimeters extrusion width = 0.45mm
6 ; perimeters extrusion width = 0.45mm
7 ; infill extrusion width = 0.45mm
8 ; solid infill extrusion width = 0.45mm
9 ; top infill extrusion width = 0.40mm
10 ; first layer extrusion width = 0.42mm
11
12 M107
13 ;TYPE:Custom
14 G90 ; use absolute coordinates
15 M83 ; extruder relative mode
16 M104 S215 ; set extruder temp
17 M140 S60 ; set bed temp
18
```

- having semicolon before the line means comment
- it is ignored

2) Gcode



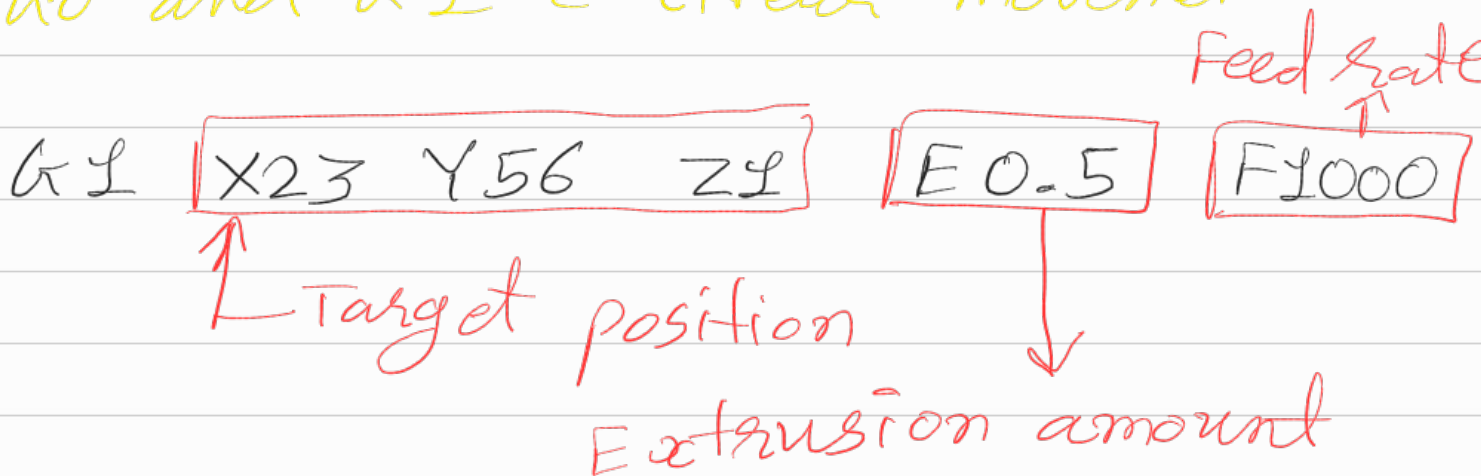
- Preparatory Commands

G0 & G1	Linear Move
G28	Homing

- Miscellaneous Commands

M104	Heat nozzle
M106	set fan speed

• G0 and G1 - linear movement



G0 used in past for rapid moves where precision wasn't needed

G1 coordinate movement

G0 — Travel move

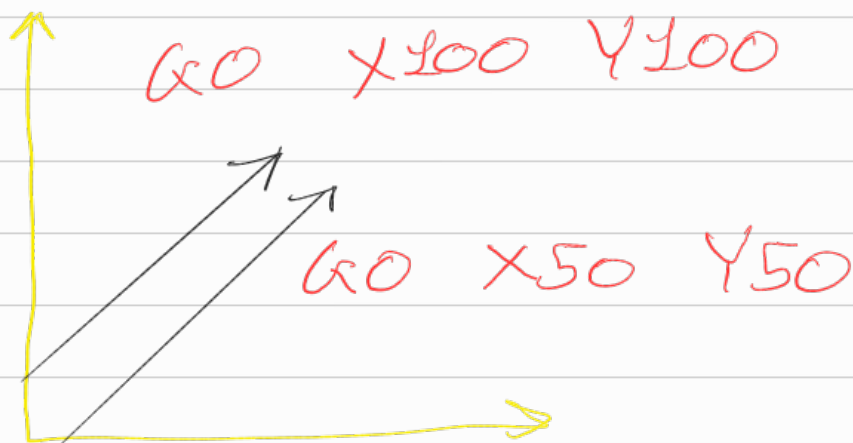
G1 — Feed move

Positioning mode

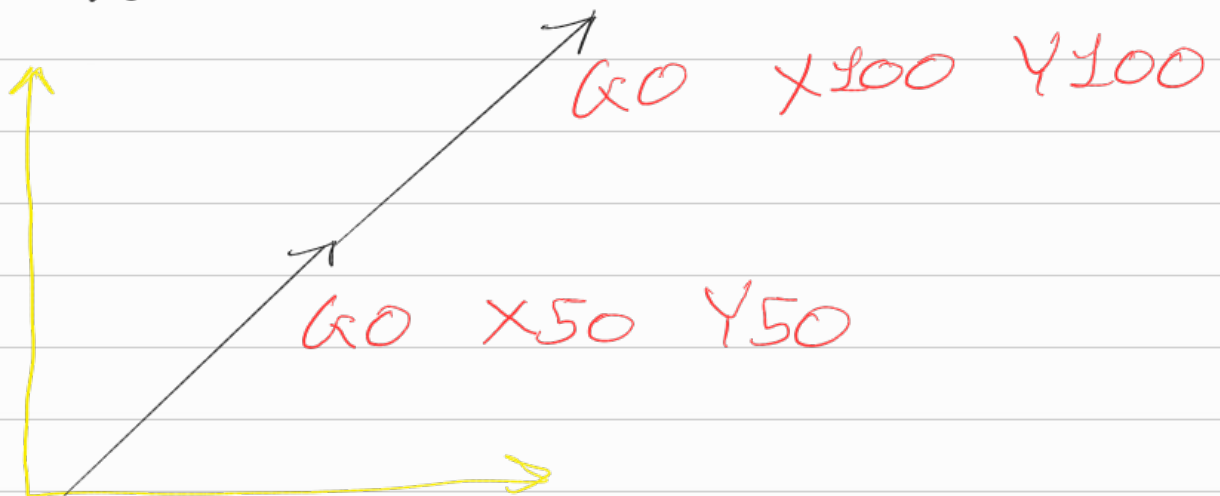
G90 — Absolute Positioning

G91 — Relative Positioning

• G90 — absolute



• G91 — Relative



G90 / G91 Use

G91 ; Relative Positioning
G0 Z10 ; Lift nozzle 10 mm
G90 ; Absolute Positioning
G0 X230 Y230 ; Move to edge

M82 and M83 extruder position
Just like G.

— G28 - Auto Homing

G28 - Home all axis

G28 X Y - Home only X and Y

G28 Z - Home only Z

- when we turn on the printer it doesn't know where it is at so that we can use G28

— G29 - For bad leveling

- G92 - set current Positioning

G92 X0 Y0 Z0 ; set all axis to zero

G92 Z0 ; set z 0.

- M104 & M109 temperature

M104 S210 ; set nozzle temperature
; continue

M109 S210 ; set nozzle temperature
; wait till the tem reaches it.

- M106 & M107 Fan speed

M106 S255 ; set full fan speed
M106 S0 ; Turn off the fan
M107 ; Turn off the fan